First-Order Models

Introduction to Logic (CS), Week 7

Dan Frumin

University of Groningen

12-13 Oct 2026

Today's Lecture

1 Overview

2 Sets, Relations, Functions

3 First-Order Models

4 First-Order Consequence and Equivalence

5 Concluding

Organisation Week 7

Lecture tomorrow:

- Another lecture by Gerard
- Exercises on Evaluating Models

Tutorials:

- Tutorials for groups gr7, gr8 scrapped, please come to a different tutorial slot

Homework Assignment 2:

- Out tomorrow after the lecture
- Deadline: one week, as usual
- Topics: FOL translations, proofs, models

This Week

This week's topics: FOL models, logical laws with quantifiers. Today we will cover the basics of set theory.

Learning goals: At the end of this week, you are able to

- construct first-order models to describe situations
- evaluate truth of sentences in first-order models
- apply logical laws to rewrite FOL sentences with quantifiers

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Talking about Sets

Today we will get introduced to the language of set theory. It is ubiquitous in anything even remotely formal (math, biology, chemistry, logic, ...).

- We will study “naive set theory” (e.g. “informal set theory”)
- Syllabus Section 8 contains a lot of material, but we will not use set operations in this course
- See also LPL Ch 1.6

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- See also LPL Ch 1.6
- LPL Ch 15 is about *axiomatic set theory* – a different beast. Ch 15 is not part of the required course readings.
- Operations on sets will be covered more in the Discrete Structures course

Sets: Unordered collections of distinct objects

Examples of sets:

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$, the set of natural numbers

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Sets are unordered, and members only occur once.

For example: $\{1, 3, 42, 3, 1, 1\} = \{1, 3, 42\}$

Elements of Sets (Membership relation $\in$)

Notation and terminology, by example:

$A = \{1, 2, 3\}$ says "A is the set containing the elements 1, 2 and 3."

$A \in B$ says "B is a set, and A is a member of of B."

(or "B is a set, and A is an element of B.")

Examples: $3 \in \{1, 2, 3\}$, $4 \notin \{1, 2, 3\}$. Nothing is a member of $\emptyset$.

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Specifying Sets

By enumeration:

- $\{1, 2, 3\}$
- $\{1, 2, 3, \ldots, 50\}$ is the set of natural numbers between 1 and 50.
- $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$ is the set of all integers.

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Let $P(x)$ be a predicate.

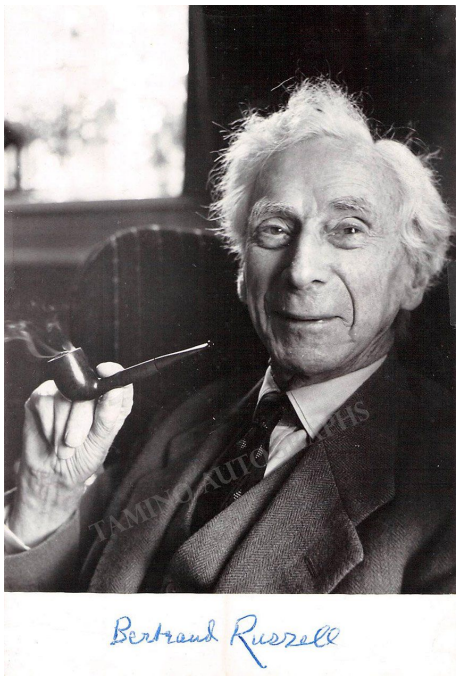
- $\{x \in A \mid P(x)\}$ (or $\{x \in A : P(x)\}$) is the set of all elements of A that satisfy P.
- $\{x \mid P(x)\}$ (or $\{x : P(x)\}$) is the set of all objects that satisfy P.

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The barber of Brașov exactly shaves the people of Brașov who do not shave themselves. Does the barber shave themselves?

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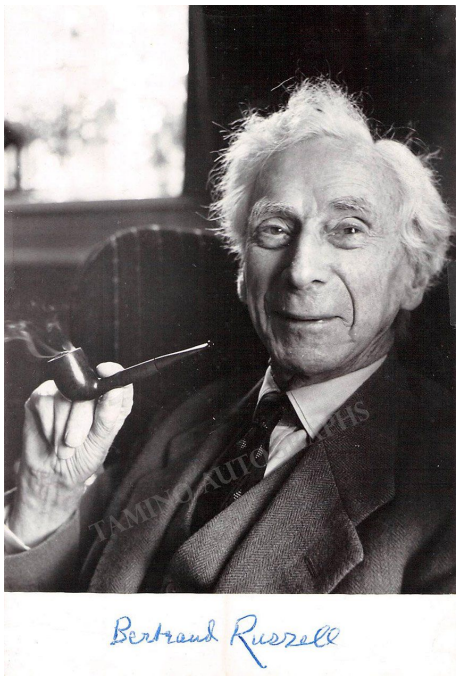
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Do we have $X \in X$?

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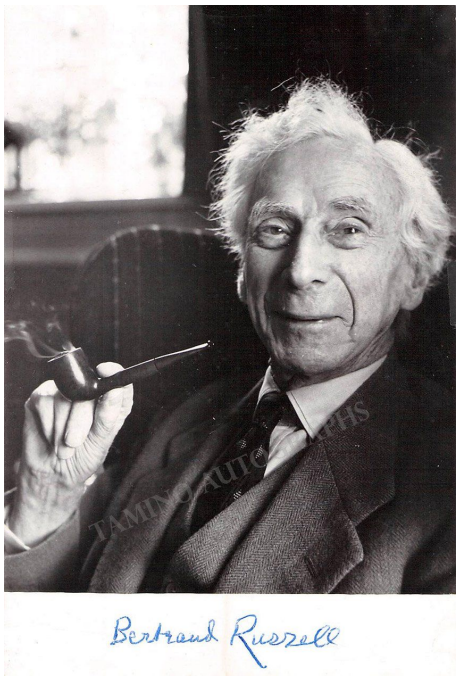
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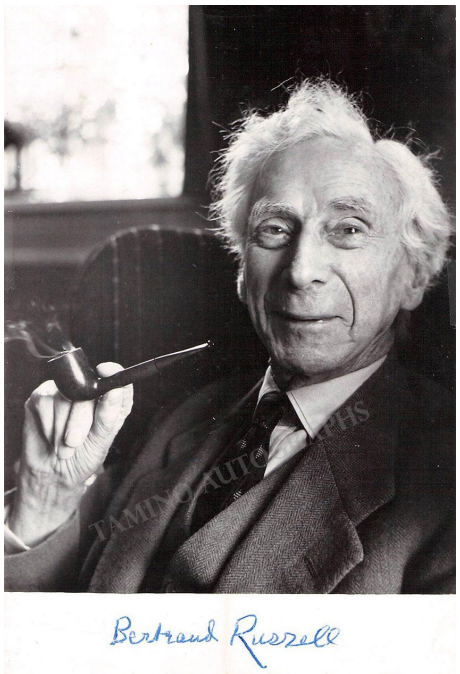
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Moral: Be careful with set abstraction. Only allow relative abstraction $\{x \in A \mid E(x)\}$ as a principle to construct sets.

Ordered Pairs

- The ordered pair of a and b is written as $\langle a, b \rangle$.
- Defining property: $\langle a, b \rangle = \langle c, d \rangle$ if and only if $a = c$ and $b = d$

So order matters: $\langle 1, 2 \rangle \neq \langle 2, 1 \rangle$.

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Generalise to:

- (ordered) n-tuples, e.g. the 5-tuple $\langle 1, 2, 2, 3, 1 \rangle$.
- Note: $\langle 1, 2, 2, 3, 1 \rangle \neq \langle 1, 1, 2, 2, 3 \rangle$ and $\langle 1, 2, 2, 3 \rangle \neq \langle 1, 2, 3 \rangle$.

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For a set A and a natural number n,

$$A^n = \{\langle a_1, \ldots, a_n \rangle \mid \text{for } i = 1, \ldots, n : a_i \in A\}$$

Example: $\mathbb{N}^3 = \{\langle a_1, a_2, a_3 \rangle \mid \text{for } i = 1, 2, 3 : a_i \in \mathbb{N}\}$

Binary Relations

- A binary relation R is a set of ordered pairs.
- An n-ary relation R is a set of n-tuples.

Examples:

- binary relations on $\mathbb{N}$: $=$, $\leq$, $\geq$, $<$, $>$.

$$\leq \quad = \quad \{\langle x, y\rangle \in \mathbb{N}^2 \mid x \text{ is less than or equal to } y\}$$

Infix-notation: For example, $x \leq 5$ instead of $\langle 3, 5\rangle \in \leq$.

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- 3-ary relation on $\mathbb{N}$: $R = \{\langle 1, 1, 2\rangle, \langle 1, 2, 2\rangle, \langle 1, 2, 1\rangle, \langle 2, 2, 2\rangle\}$

Note: We use the symbol $=$ to denote the equality predicate between relations, sets, numbers, ...

Properties of Binary Relations

Let R be a binary relation on some set A. R may satisfy certain properties:

reflexivity for all $x \in A$: $\langle x, x \rangle \in R$

symmetry for all $x, y \in A$: if $\langle x, y \rangle \in R$, then $\langle y, x \rangle \in R$.

transitivity for all $x, y, z \in A$: if $\langle x, y \rangle \in R$ and $\langle y, z \rangle \in R$, then $\langle x, z \rangle \in R$.

Functions

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$$f : A \to B$$

map elements of the set A to the set B.
E.g. $f(x) = x + 1$ is a function $f : \mathbb{N} \to \mathbb{N}$.

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Functions on multiple arguments are functions on tuples: $f : \mathbb{N}^2 \to \mathbb{N}$

$$f(\langle a, b \rangle) = a + 2b$$

Functions as special relations

We can encode a function $f : \mathbb{N} \to \mathbb{N}$ as a binary relation

$$\{\langle 0, 1\rangle, \langle 1, 2\rangle, \langle 2, 3\rangle, \dots\}$$

In each tuple $\langle a, b\rangle$ the first component is the input, and the second is the output of the function.

Functions as special relations

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Functions can be seen as *special relations*

Functions as special relations

Let R be a binary relation on a set A.
We say R is a unary functional if

- for all $x, y, z \in A$:
  if $\langle x, y\rangle \in R$ and $\langle x, z\rangle \in R$, then $y = z$
  (inputs are assigned unique outputs)

We say that R is a total functional, if R is a function and

- for all $a \in A$, there is a $b \in A$ such that $\langle a, b\rangle \in R$
  (an output is defined for all inputs)

In the later case R represents the function $f : A \to A$ such that $f(a) = b$ iff $\langle a, b\rangle \in R$.

Functional relations and domains

A *functional relation R* that is non-total still defines a function on particular *domain*.

E.g. $\{\langle 0,1\rangle, \langle 1,0\rangle\}$ defines a function $f : \{0,1\} \to \mathbb{N}$,
but not a function $f : \mathbb{N} \to \mathbb{N}$.

n-ary Functions

Let R be a $(n+1)$-ary relation on a set A.
We say R is an n-ary function if

- for all $\langle a_1, \ldots, a_n \rangle \in A^n$ and $a, b \in A$
  if $\langle a_1, \ldots, a_n, a \rangle, \langle a_1, \ldots, a_n, b \rangle \in R$ then $a = b$.
  (inputs from A^n are assigned unique outputs)

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- for all $\langle a_1, \ldots, a_n \rangle \in A^n$ there is an $a \in A$ such that
  $\langle a_1, \ldots, a_n, a \rangle \in R$.

  (an output is defined for all inputs from A^n)

Quick Quiz

Which of the following relations represent functions?

a. $\{\langle x, y\rangle \in \mathbb{N}^2 \mid x \leq 2y\}$
b. $\{\langle 0, 0\rangle, \langle 1, 1\rangle, \langle 2, 4\rangle, \langle 3, 9\rangle, \langle 4, 16\rangle, ...\}$
c. $\{\langle 0, 0\rangle, \langle 1, 10\rangle, \langle 3, 30\rangle\}$
d. $\{\langle 0, 0\rangle, \langle 10, 1\rangle, \langle 30, 3\rangle\}$
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Answer: b,c,d

Encoding tuples as set

Set-theoretic hack: we can encode ordered tuples as unordered sets!

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In particular

$$\langle a, b \rangle = \{a, \{b\}\} \neq \{b, \{a\}\} = \langle b, a \rangle$$

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- 0 as $\emptyset$
- 1 as $\{0\} = \{\emptyset\}$
- 2 as $\{0, 1\} = \{\emptyset, \{\emptyset\}\}$
- ...
- $n + 1$ as $\{0, \ldots, n\}$

Then $n \leq m$ just means $n \in m$.

We are not going to do this encoding! :) Look up “Von Neumann numerals” if you are interested.

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- Recall: Logical consequence for sentences of PL can be determined using truth tables.
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- Recall: Logical consequence for sentences of PL can be determined using truth tables.
- Quantifiers can not be described with truth tables
- We want: a mathematically precise definition of “situations” for full FOL. (TT row fully describes “a situation” for PL)

$$\frac{\text{truth tables}}{\text{PL}} \qquad \frac{\text{first-order models}}{\text{FOL}}$$

(first-order model = first-order structure)

First-Order Models (Intuition)

How should a first-order model look like? What is the minimum amount of information we need a) formalize a given scenario, b) evaluate whether a sentence is true in a given scenario?

All cubes are pink.

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   we want set C and P of elements from D, e.g. $\{a, b\}$ and $\{b, c\}$
3. To check the sentence is to check:
   for any object $x \in \{a, b, c\}$, if $x \in \{a, b\}$ then $x \in \{b, c\}$
   is this true?

First-Order Model for a Language $\mathcal{L}$ (by example)

Vocabulary of $\mathcal{L}$ (syntax):

- binary predicate $Likes(x, y)$: *x likes y*
- constants: a, b, c
- unary function symbol $bf(x)$: *the best friend of x*

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A model formally describes a *situation* or a scenario expressible in the language $\mathcal{L}$.

First-order models: Example

The model:

The situation:

- We have a population of students, and we are interested in three of them in particular: Anne, Benjamin, and Cornelius.
- Benjamin is the best friend of Anne and Cornelius, and Anne is the best friend of Benjamin.
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- $\mathfrak{M}(Likes) = \{\langle Anne, Benjamin\rangle, \dots\}$

First-order models: Example cont.

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- $\mathfrak{M}(Likes) = \{ \begin{array}{l} \langle Anne, Benjamin\rangle, \langle Benjamin, Anne\rangle, \\ \langle Cornelius, Benjamin\rangle, \langle Benjamin, Cornelius\rangle \end{array} \}$

At its core the model describes basic facts about the situation:

$$\mathfrak{M} \models Likes(c, b) \qquad \mathfrak{M} \not\models Likes(c, c) \qquad \mathfrak{M} \models Likes(bf(a), c)$$

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From basic facts we can form and evaluate compound sentences about the scenario:

$$\mathfrak{M} \models Likes(c, b) \wedge \neg Likes(c, c) \qquad \mathfrak{M} \models Likes(c, c) \rightarrow Likes(b, b)$$

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Including sentences with quantifiers:

$$\mathfrak{M} \models \exists x(Likes(x, a)) \qquad \mathfrak{M} \models \forall x \forall y(Likes(x, y) \rightarrow Likes(y, x))$$

FOL Models and Evaluation

How do we describe models in general, and how do we evaluate arbitrary formulas in a model?

Tricky bit: quantifiers and variable assignments

FOL Models in General

A model is always given relative to a vocabulary $\mathcal{L}$! An $\mathcal{L}$-model (or, just a model) $\mathfrak{M}$ consists of

- Non-empty domain of discourse $\mathfrak{M}(\forall) = D$
- For each constant a in $\mathcal{L}$ an interpretation $\mathfrak{M}(a) \in D$
- For each function symbol f of arity n a function $\mathfrak{M}(f) : D^n \to D$
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We want to evaluate whether a complex formula P is true or not in the model

$$\mathfrak{M} \models P \qquad \mathfrak{M} \not\models P$$

Compositionality of semantics

We want to determine the truth of a formula P in terms of truth-values of its constituents.

E.g.

$$\mathfrak{M} \models P \wedge Q \quad \text{iff} \quad \mathfrak{M} \models P \text{ and } \mathfrak{M} \models Q$$

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$$\mathfrak{M} \models \forall x(P(x)) \quad \text{iff} \quad ???\mathfrak{M} \models P(x)???$$

Variable Assignments

- We want to define when a sentence is true in a model $\mathfrak{M}$.
- Recall (from week 5): $\forall x\, P(x)$ is true if every object in D satisfies $P(x)$. So we need to consider formulas with variables
- Previously, we introduced temporary names for objects. Now we make this mathematically precise.

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- E.g. h with $h(x) = \mathit{Anne}$ we interpret $\mathit{Likes}(\mathit{bf}(b), x)$ as “The best friend of Benjamin likes Anne”.

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- When interpreting quantifiers we will need to *modify* or *update* the variable assignment.
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$$\begin{aligned} h[x/d](x) &= d \\ h[x/d](y) &= h(y) \quad \text{if } y \neq x \end{aligned}$$

Variable Assignments: Example

Suppose:
$D = \mathbb{N}$ and VAR = $\{x, y, z, \ldots\}$,
and h is the partial function:

$$\begin{aligned} h(x) &= 42 \\ h(y) &= 11 \end{aligned}$$

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E.g. $[\![bf(x)]\!]^{\mathfrak{M}}_h = \textit{Benjamin}$ with $h(x) = \textit{Anne}$

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A variable assignment h is appropriate for a term t if it is defined for all variables occurring in t.

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Continue our example. Recall:

$$\begin{aligned} \mathfrak{M}(bf)(Anne) &= Benjamin, \\ \mathfrak{M}(bf)(Benjamin) &= Anne, \\ \mathfrak{M}(bf)(Cornelius) &= Benjamin. \end{aligned}$$

Suppose variable assignment h is such that $h(x) = Anne$.

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Satisfaction of Formulas in a Model

Let A be a wff, $\mathfrak{M}$ a model, and h a variable assignment appropriate for A. We write $\mathfrak{M} \models A[h]$ to say that "h satisfies A in $\mathfrak{M}$" or "$\mathfrak{M}$ satisfies A under h":

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Satisfaction of Formulas in a Model

Let A be a wff, $\mathfrak{M}$ a model, and h a variable assignment appropriate for A. We write $\mathfrak{M} \models A[h]$ to say that "h satisfies A in $\mathfrak{M}$" or "$\mathfrak{M}$ satisfies A under h":

$$\mathfrak{M} \models R(t_1, \ldots t_n)[h] \quad \text{iff} \quad \langle [\![t_1]\!]^{\mathfrak{M}}_h, \ldots, [\![t_n]\!]^{\mathfrak{M}}_h \rangle \in \mathfrak{M}(R)$$

$$\mathfrak{M} \models \neg A[h] \quad \text{iff} \quad \mathfrak{M} \not\models A[h] \text{ (i.e. we do not have } \mathfrak{M} \models A[h])$$

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$\mathfrak{M} \models \neg A[h]$	iff	$\mathfrak{M} \not\models A[h]$ (i.e. we do not have $\mathfrak{M} \models A[h]$)
$\mathfrak{M} \models (A \wedge B)[h]$	iff	$\mathfrak{M} \models A[h]$ and $\mathfrak{M} \models B[h]$
$\mathfrak{M} \models (A \vee B)[h]$	iff	$\mathfrak{M} \models A[h]$ or $\mathfrak{M} \models B[h]$ (or both)
$\mathfrak{M} \models (A \rightarrow B)[h]$	iff	$\mathfrak{M} \not\models A[h]$ or $\mathfrak{M} \models B[h]$ (or both)
$\mathfrak{M} \models (A \leftrightarrow B)[h]$	iff	($\mathfrak{M} \models A[h]$ and $\mathfrak{M} \models B[h]$) or ($\mathfrak{M} \not\models A[h]$ and $\mathfrak{M} \not\models B[h]$)

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$$\mathfrak{M} \models \forall x A[h] \quad \text{iff} \quad \text{for all } d \in \mathfrak{M}(\forall) : \mathfrak{M} \models A[h[x/d]]$$

Satisfaction of Formulas in a Model

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$$\mathfrak{M} \models \exists x A[h] \quad \text{iff} \quad \text{for some } d \in \mathfrak{M}(\forall) : \mathfrak{M} \models A[h[x/d]]$$

Object-Notation vs Meta-Notation

Be careful to not mix them up!

Object-notation:

- Syntax that is part of wffs in some FOL language $\mathcal{L}$.
- Example: $\neg R(x,x) \wedge \forall z(R(x,z) \vee P(z))$

Meta-notation:

- Logical statements at the meta-level in English (not a wff).
- Example:

  Good: $\mathfrak{M} \models (A \wedge B)[h]$ iff $\mathfrak{M} \models A[h]$ and $\mathfrak{M} \models B[h]$.

  Bad: $\mathfrak{M} \models (A \wedge B)[h]$ iff $\mathfrak{M} \models A[h] \wedge \mathfrak{M} \models B[h]$.

  ($\wedge$ can only be applied to wffs.)

First-Order Satisfaction: Example 1

Continue our small example. Let $A = \exists y(\mathit{Likes}(x, y) \wedge \mathit{Likes}(y, x))$. Let $h(x) = \mathit{Benjamin}$. We claim that: $\mathfrak{M} \models A[h]$.

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We claim that we can take $d = Anne$. We have:

$$\mathfrak{M} \models (Likes(x, y) \wedge Likes(y, x))[h[y/Anne]]$$

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$\mathfrak{M} \models (Likes(x, y) \wedge Likes(y, x))[h[y/Anne]]$

iff $\mathfrak{M} \models Likes(x, y)[h[y/Anne]]$ and $\mathfrak{M} \models Likes(y, x)[h[y/Anne]]$

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iff $\mathfrak{M} \models Likes(x, y)[h[y/Anne]]$ and $\mathfrak{M} \models Likes(y, x)[h[y/Anne]]$

iff $\langle [\![x]\!]^{\mathfrak{M}}_{h[y/Anne]}, [\![y]\!]^{\mathfrak{M}}_{h[y/Anne]} \rangle \in \mathfrak{M}(Likes)$ and
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iff $\langle Anne, Benjamin \rangle \in \mathfrak{M}(Likes)$ and $\langle Benjamin, Anne \rangle \in \mathfrak{M}(Likes)$

iff True and True

iff True

First-Order Truth of Sentences

Let S be an $\mathcal{L}$-sentence, i.e. a formula of first-order language $\mathcal{L}$ with no free variables, and let $\mathfrak{M}$ be an $\mathcal{L}$-model.

- Note: Since S is a sentence, any variable assignment is appropriate for S, including the empty assignment (the partial function undefined for all variables).

- **Def.** S is true in $\mathfrak{M}$ iff the empty variable assignment satisfies S in $\mathfrak{M}$. When this is the case, we write $\mathfrak{M} \models S$ (i.e. no variable assignment).

First-Order Truth of Sentences: Example 2

Consider the same model and a sentence $\forall x(\mathit{Likes}(x, \mathit{bf}(\mathit{bf}(x))))$
We claim that: $\mathfrak{M} \not\models \forall x(\mathit{Likes}(x, \mathit{bf}(\mathit{bf}(x))))$

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It is *not* the case that $\mathfrak{M} \models \forall x(\mathit{Likes}(x, \mathit{bf}(\mathit{bf}(x))))$

It is *not* the case that for all $d \in \mathfrak{M}(\forall), \mathfrak{M} \models (\mathit{Likes}(x, \mathit{bf}(\mathit{bf}(x))))h[d/x]$

It is *not* the case that for all $d \in \mathfrak{M}(\forall), \langle d, \mathfrak{M}(\mathit{bf})(\mathfrak{M}(\mathit{bf})(d))\rangle \in \mathfrak{M}(\mathit{Likes})$

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We need to find a $d \in \mathfrak{M}(\forall)$ such that
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We need to find a $d \in \mathfrak{M}(\forall)$ such that
$\langle d, \mathfrak{M}(\mathit{bf})(\mathfrak{M}(\mathit{bf})(d))\rangle \notin \mathfrak{M}(\mathit{Likes})$. E.g. $d = \mathit{Anne}$ suffices!

Another example (in the same model): $\forall x \exists y(\mathit{Likes}(x, \mathit{bf}(y)))$.
Does the model satisfy this sentence?

Another example (in the same model): $\forall x \exists y(\mathit{Likes}(x, \mathit{bf}(y)))$.
Does the model satisfy this sentence?

Consider the case when we take x to be Benjamin.

First-Order Models

More examples on FO models in lecture tomorrow.
And in the file: `semanticsWorkout.pdf`
(attached to Week 7 info).

Today's Lecture

1 Overview

2 Sets, Relations, Functions

3 First-Order Models

4 First-Order Consequence and Equivalence

5 Concluding

First-Order Consequence, Validity, Satisfiability

Let $\mathcal{L}$ be a first-order language.

- An $\mathcal{L}$-sentence S is a first-order consequence of a sentence P if for any $\mathcal{L}$-model $\mathfrak{M}$: if $\mathfrak{M}$ validates P then $\mathfrak{M}$ validates S.

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- An $\mathcal{L}$-sentence S is a first-order consequence of a set $\mathcal{T} = \{T_1, T_2, \ldots\}$ of $\mathcal{L}$-sentences if for all $\mathcal{L}$-models $\mathfrak{M}$:

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  (cf. validity of arguments.)

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  (cf. validity of arguments.)
- Note: To show that $\mathcal{T} \not\models S$, give a countermodel, i.e., an $\mathcal{L}$-model $\mathfrak{M}$ such that for all T_i, $\mathfrak{M} \models T_i$, but $\mathfrak{M} \not\models S$.

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- An $\mathcal{L}$-sentence S is a first-order validity (or: S is *valid*) if $\mathfrak{M} \models S$ for *all* $\mathcal{L}$-models $\mathfrak{M}$. (We write $\models S$).

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- An $\mathcal{L}$-sentence S is a first-order validity (or: S is *valid*) if $\mathfrak{M} \models S$ for *all* $\mathcal{L}$-models $\mathfrak{M}$. (We write $\models S$).
- An $\mathcal{L}$-sentence S is first-order satisfiable if $\mathfrak{M} \models S$ for *some* $\mathcal{L}$-model $\mathfrak{M}$.

Provability vs Validity

We have two important notions.

Let S be a sentence of FOL.

- S is provable in System $\mathcal{F}$ (i.e. there is a formal proof for S);
- S is valid, i.e., S is true in all models.

How are these notions related?

Soundness and completeness

Theorem.
We have for all FOL sentences $\mathcal{T} = \{T_1, \ldots, T_n\}, S$:

$$\mathcal{T} \vdash S \text{ is provable in System } \mathcal{F} \quad \text{if and only if} \quad \mathcal{T} \models S.$$

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Soundness of System $\mathcal{F}$: "provable $\Rightarrow$ valid"
(fairly easy, see LPL Ch 18.3, not part of course material)

Completeness of System $\mathcal{F}$: "valid $\Rightarrow$ provable"
(rather difficult, more advanced logic)

Remark:
To show that argument $\mathcal{T} \vdash S$ is valid: give a proof.
To show that argument $\mathcal{T} \vdash S$ is not valid: give countermodel.

First-Order Equivalence

Assume we have fixed a FO language $\mathcal{L}$.

Definition.

- Two FOL sentences A and B are FO equivalent (notation: $A \Leftrightarrow B$) if for all FO models $\mathfrak{M}$:

$$\mathfrak{M} \models A \quad \text{iff} \quad \mathfrak{M} \models B$$

- Two FOL wffs A and B are FO equivalent (notation: $A \Leftrightarrow B$) if for all FO models $\mathfrak{M}$ and all variable assignments h:

$$\mathfrak{M} \models A[h] \quad \text{iff} \quad \mathfrak{M} \models B[h]$$

Note (relation to FO consequence): $A \Leftrightarrow B$ holds iff $A \models B$ and $B \models A$, which we will also write as $A \Rightarrow B$ and $B \Rightarrow A$, respectively.

Substituting Equivalent Wffs

The principle of substitution applies to FOL:

Principle of Substitution of Equivalents (FOL)

Let P and Q be wffs. For all sentences $S(P)$,

$$\text{if} \quad P \Leftrightarrow Q \quad \text{then} \quad S(P) \Leftrightarrow S(Q).$$

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Example: Since $Cube(x) \rightarrow Small(x) \Leftrightarrow \neg Cube(x) \vee Small(x)$

$$\forall x(Cube(x) \rightarrow Small(x)) \quad \Leftrightarrow \quad \forall x(\neg Cube(x) \vee Small(x))$$

But there are logical laws that involve quantifiers in an essential way.

Null Quantification

What to think of $\forall xA$ and $\exists xA$ if x is not in A?

$\forall x\ (2+2=4)$ true, because $2+2=4$

$\forall x\ (2+2=5)$ false, because $2+2 \neq 5$

$\exists x\ (2+2=4)$ true, because $2+2=4$

$\exists x\ (2+2=5)$ false, because $2+2 \neq 5$

$\exists x\ \mathit{Cube}(b)$ true iff $\mathit{Cube}(b)$ true

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$\exists x\ \mathit{Cube}(b)$ true iff $\mathit{Cube}(b)$ true

So we have

if x is not in A, then $\quad \forall xA \Leftrightarrow \exists xA \Leftrightarrow A$

Renaming Bound Variables

The name of a bound variable is not relevant: we have

$$\boxed{\forall x A(x) \Leftrightarrow \forall y A(y)} \qquad \boxed{\exists x A(x) \Leftrightarrow \exists y A(y)}$$

Example:

$$\exists x (P(x) \wedge \forall z\, R(x, z)) \Leftrightarrow \exists y (P(y) \wedge \forall z\, R(y, z))$$

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Example:

$$\exists x(P(x) \wedge \forall z\, R(x,z)) \Leftrightarrow \exists y(P(y) \wedge \forall z\, R(y,z))$$

Use these laws to avoid *variable clashes* such as

$$\exists x(f(x) > 5 \wedge \forall x(g(x) = 7))$$

and write instead

$$\exists y(f(y) > 5 \wedge \forall x(g(x) = 7))$$

De Morgan for Quantifiers

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We have:

$$\boxed{\neg\forall x\, A(x) \;\Leftrightarrow\; \exists x \neg A(x)} \qquad \text{recall: } \neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$$

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It follows that $\forall$ and $\exists$ are interdefinable:

$$\exists x\, A(x) \;\Leftrightarrow\; \neg\forall x \neg A(x) \qquad \text{recall: } A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$$
$$\forall x\, A(x) \;\Leftrightarrow\; \neg\exists x \neg A(x) \qquad \text{recall: } A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$$

Interaction of $\forall$ with $\wedge$ and $\vee$

- We have: $\boxed{\forall x(A(x) \wedge B(x)) \Leftrightarrow \forall x\, A(x) \wedge \forall x B(x)}$

  Example: $\forall x(\mathit{Small}(x) \wedge \mathit{Cube}(x)) \Leftrightarrow \forall x\, \mathit{Small}(x) \wedge \forall x\, \mathit{Cube}(x)$

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Combining Quantifiers

When combining quantifiers of the same kind, order does not matter:

$$\forall x \forall y\, A(x, y) \;\Leftrightarrow\; \forall y \forall x\, A(x, y)$$

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When combining quantifiers of different kinds, order matters if x and y are related in A:

$$\forall x \exists y\, A(x,y) \;\not\Rightarrow\; \exists y \forall x\, A(x,y)$$

$$\exists x \forall y\, A(x,y) \;\Rightarrow\; \forall y \exists x\, A(x,y)$$

Think of: $\forall x \exists y (x < y)$ vs $\forall x \exists y (\mathit{Even}(x) \wedge \mathit{Prime}(y))$.

Today's Lecture

1 Overview

2 Sets, Relations, Functions

3 First-Order Models

4 First-Order Consequence and Equivalence

5 Concluding

Wrapping up

Learning goals: At the end of this week, you are able to

- evaluate truth of sentences in first-order models
- apply logical laws to rewrite FOL sentences (with quantifiers)

Next (last!) week:

- Prenex and Skolem normal form
- Satisfiability and Horn sentences
- Applications of logic in CS